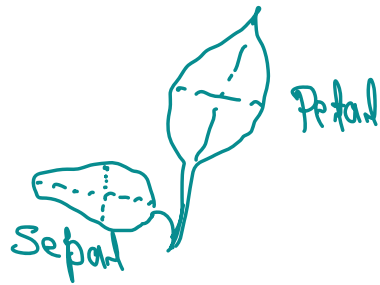


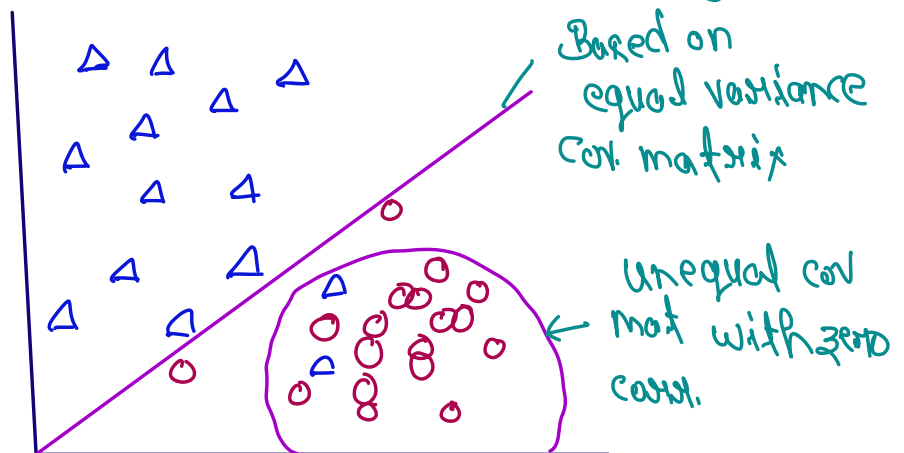
Example 1: Fisher 1936

Two Species: 1) versicolour
2) virginica



50 obs. for each species. Given a data (width and length of sepal and petal) of a new species, where will you classify?

Example 2: Smith (1947): Smith 1947, collected information on normal individual and psychotic individuals.



Formulation: Consider g populations or groups $\pi_1, \dots, \pi_g$; $g \geq 2$. Associated with each population π_j there is a probability density $f_j(x)$ on $\mathbb{R}^p$. That is if an individual come from a population π_j , he has the p.d.f. $f_j(x)$. The objective of discriminant analysis is to allocate an individual to one of those g groups on the basis of his measurements x .

Discriminant Rule: A discriminant rule d corresponds to a division of $\mathbb{R}^p$ into disjoint regions $R_1, \dots, R_g$ ($\bigcup_{i=1}^g R_i = \mathbb{R}^p$). The rule d is defined by

allocate x to π_j if $x \in R_j$

for $j = 1, \dots, g$.

Discrimination will be more accurate if π_j has most of its probability concentrated in R_j for each j .

The situation where p.d.f. $f_j(x)$ are known exactly is the simplest to analyse theoretically, although it is the least realistic in practice.

A variant of this situation occurs when the form of the p.d.f. for each population is known, but there are parameters which must be estimated. The estimation is based on a sample data matrix $X(n \times p)$ whose rows are partitioned into g groups.

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_g \end{bmatrix}_{n \times p}$$

Explain it using
previous example

The $n_j \times p$ matrix x_j represents a sample of n_j individuals from the popⁿ π_j .

Discrimination when the Population are Known:

The maximum likelihood discrimination rule:

Consider the situation where the exact distⁿ of the pop^s $\pi_1, \dots, \pi_g$ are known. Write the p.d.f. of the j th popⁿ as $f_j(X) = L_j(X)$.

Definition: The "maximum likelihood discrimination rule" for allocating an observation $\underline{x}$ to one of the populations $\pi_1, \dots, \pi_g$ is to allocate $\underline{x}$ to the population which gives the largest likelihood to $\underline{x}$.

That is, the maximum likelihood rule says one should allocate $\underline{x}$ to π_j , where

$$L_j(X) = \max_i L_i(X).$$

If several likelihoods take the same maximum value, then any of these may be chosen.

Example 1: Let x is a 0-1 scaled random variable
 and if π_1 is the popⁿ with probs $(\frac{1}{2}, \frac{1}{2})$ and
 π_2 is the popⁿ with probs $(\frac{1}{4}, \frac{3}{4})$, then the
 maximum likelihood discrimination rule allocates x to
 π_1 when $x=0$ and allocates x to π_2 when $x=1$.
 This is because $L_1(0) = \frac{1}{2} > L_2(0) = \frac{1}{4}$, and
 $L_2(1) = \frac{3}{4} > L_1(1) = \frac{1}{2}$.

Example 2: Suppose that $\underline{x}$ is multinomial random
 vector, which comes either from π_1 with
 multinomial probabilities $\alpha_1, \dots, \alpha_k$ or from π_2 ,
 with multinomial probs $\beta_1, \dots, \beta_k$ where $\sum \alpha_i = \sum \beta_i = 1$,
 and $\sum \alpha_i = n$, fixed.

For $\underline{x}$ in π_1 , its likelihood is

$$\frac{n!}{x_1! \dots x_k!} \alpha_1^{x_1} \dots \alpha_k^{x_k}.$$

If $\underline{x}$ comes from π_2 the likelihood is

$$\frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}.$$

If λ is the ratio of these likelihoods, then the log likelihood ratio is

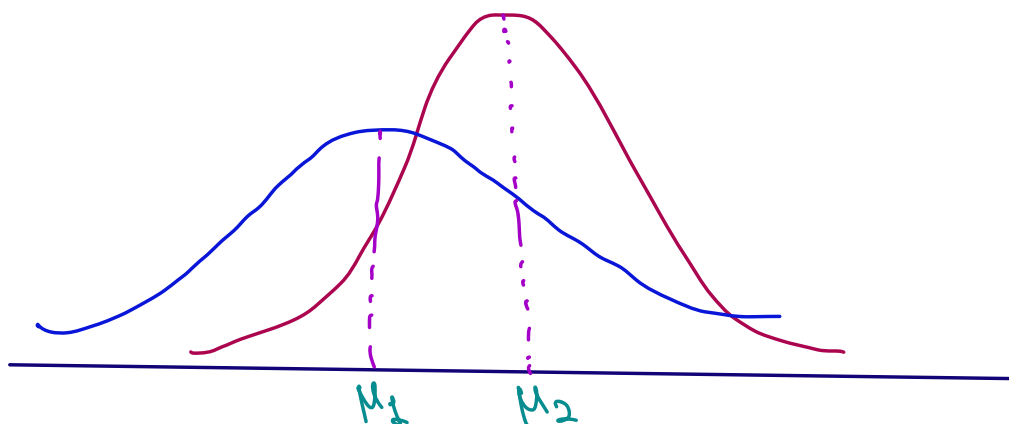
$$\log \lambda = \sum x_i \log \frac{\alpha_i}{\beta_i} = \sum x_i s_i,$$

where $s_i = \log(\alpha_i/\beta_i)$.

The maximum likelihood discriminant rule allocates $\underline{x}$ to π_1 if $\lambda > 1$, i.e. if $\log \lambda > 0$.

Example 3: Suppose $\pi_1 \sim N(\mu_1, \sigma_1^2)$ and $\pi_2 \sim N(\mu_2, \sigma_2^2)$.

Let $\mu_2 > \mu_1$ and $\sigma_1 > \sigma_2$.



The likelihood L_i is

$$L_i(x) = (2\pi\sigma_i^2)^{-1/2} \exp\left\{-\frac{1}{2} \frac{(x-\mu_i)^2}{\sigma_i^2}\right\}.$$

$L_1(x)$ exceeds $L_2(x)$ if

$$\frac{\sigma_2}{\sigma_1} \exp\left\{-\frac{1}{2} \left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \left(\frac{x-\mu_2}{\sigma_2}\right)^2\right]\right\} > 1$$

on taking the logarithms and rearranging, this inequality becomes

$$x^2 \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2}\right) - 2x \left(\frac{\mu_1}{\sigma_1^2} - \frac{\mu_2}{\sigma_2^2}\right) + \left(\frac{\mu_1^2}{\sigma_1^2} - \frac{\mu_2^2}{\sigma_2^2}\right) < 2 \log \frac{\sigma_2}{\sigma_1}.$$

The set of x 's for which the above inequality is satisfied falls in two distinct regions, one having low values of x and other having high values of x . (Home work!).

Example 4: An important special case of the previous example occurs when $\sigma_1 = \sigma_2$.

Then

$$-2\alpha \cdot \frac{(M_1 - M_2)}{\sigma^2} + \frac{(M_1 - M_2)(M_1 + M_2)}{\sigma^2} < 0$$

$$\Rightarrow (M_1 - M_2)(-2\alpha + (M_1 + M_2)) < 0$$

$$\text{If } M_2 > M_1 \text{ then } -2\alpha + (M_1 + M_2) > 0$$

$$\Rightarrow \frac{M_1 + M_2}{2} > \alpha \text{ to allocate to } \pi_1.$$

Theorem: D1

a) If π_i is the $N_p(\underline{\mu}_i, \Sigma)$ population, $i=1, \dots, g$, and $\Sigma > 0$, then the maximum likelihood discriminant rule allocates $\underline{x}$ to π_j where $j \in \{1, \dots, g\}$ is that value of i which maximizes the square of the Mahalanobis distance

$$(\underline{x} - \underline{\mu}_i)^T \Sigma^{-1} (\underline{x} - \underline{\mu}_i).$$

b) When $g=2$, the rule allocates x to π_1 if

$$\alpha^T (x - \mu) > 0 \quad \text{--- } \textcircled{D}$$

where $\alpha = \Sigma^{-1}(\mu_1 - \mu_2)$ and $\mu = \frac{1}{2}(\mu_1 + \mu_2)$
and to π_2 otherwise.

Proof: The i -th likelihood is

$$L_i(x) = |\alpha \pi \Sigma|^{-1/k} \exp\left\{-\frac{(x - \mu)^T \Sigma^{-1} (x - \mu)}{2}\right\}$$

This is maximized when the exponent is minimized, which proves part (a).

For part (b) note that $L_1(x) > L_2(x)$ if and only if

$$(\underline{x} - \underline{\mu}_1)^T \Sigma^{-1} (\underline{x} - \underline{\mu}_1) < (\underline{x} - \underline{\mu}_2)^T \Sigma^{-1} (\underline{x} - \underline{\mu}_2)$$

$$\begin{aligned} \Rightarrow -\underline{\mu}_1^T \Sigma^{-1} \underline{x} - \underline{x}^T \Sigma^{-1} \underline{\mu}_1 + \underline{\mu}_1^T \Sigma^{-1} \underline{\mu}_1 \\ < -\underline{\mu}_2^T \Sigma^{-1} \underline{x} - \underline{x}^T \Sigma^{-1} \underline{\mu}_2 + \underline{\mu}_2^T \Sigma^{-1} \underline{\mu}_2 \end{aligned}$$

Claim: $\alpha^T (x - \mu) > 0$

$$(\underline{\mu}_1^T - \underline{\mu}_2^T) \Sigma^{-1} \left(\underline{x} - \frac{\underline{\mu}_1}{2} - \frac{\underline{\mu}_2}{2} \right) > 0$$

$$\Rightarrow \underline{\mu}_1^T \Sigma^{-1} \underline{x} - \underline{\mu}_2^T \Sigma^{-1} \underline{x} - \frac{\underline{\mu}_1^T \Sigma^{-1} \underline{\mu}_1}{2} + \frac{\underline{\mu}_2^T \Sigma^{-1} \underline{\mu}_2}{2}$$

$$\rightarrow \frac{M_2^T \Sigma^{-1} M_1}{2} - \frac{M_1^T \Sigma^{-1} M_2}{2}$$

Note that when there are just $g=2$ groups, the ML discrimination rule is defined in terms of the discrimination function

$$R(x) = \log L_1(x) - \log L_2(x) \quad - (2)$$

and the ML rule takes the form

$$\begin{aligned} &\text{allocate } x \text{ to } \pi_1 \text{ if } R(x) > 0 \\ &" \quad " \quad \pi_2 \text{ if } R(x) < 0. \end{aligned}$$

The Bayes discriminant rule: In certain situations it makes sense to suppose that the various pops have "prior probabilities".

Defⁿ: If pops $\pi_1, \dots, \pi_g$ have prior prob.

$(\pi_1, \dots, \pi_g) = \underline{\pi}^T$, then the Bayes discriminant rule (with respect to $\underline{\pi}$) allocates an observation x to popⁿ for which

$$\pi_j L_j(x)$$

is maximized.

(3)

For $g=2$,

$$\begin{aligned} R(x) &= \log\{\pi_1 L_1(x)\} - \log\{\pi_2 L_2(x)\} \\ &= \log(\pi_1/\pi_2) + \log L_1(x) - \log L_2(x) \\ &= \log(\pi_1/\pi_2) + R(x) \end{aligned}$$

$$R(x) > 0 \Rightarrow R(x) > \log(\pi_2/\pi_1)$$

$\Rightarrow$ allocate to π_1 otherwise
to π_2 .

Defⁿ: Randomized discrimination rule:

A randomized discrimination rule d involves allocating an observation $\underline{x}$ to a popⁿ j with prob. $\phi_j(\underline{x})$, where $\phi_1, \dots, \phi_g$ are non-negative functions each defined on $\mathbb{R}^p$, and satisfy $\sum \phi_j(\underline{x}) = 1$ for all $\underline{x}$.

Deterministic allocation rule is a special case of a randomized allocation rule obtained by putting $\phi_j(\underline{x}) = 1$ for $\underline{x} \in R_j$ and $\phi_j(\underline{x}) = 0$ elsewhere.

For example Bayes rule with respect to prior prob. $\pi_1, \dots, \pi_g$ is defined by

$$\phi_j(x) = \begin{cases} 1 & \text{if } \pi_j L_j(x) = \max_i \pi_i L_i(x) \\ 0 & \text{o/w,} \end{cases} \quad \text{--- (3)}$$

The prob. of allocating an individual to popⁿ π_i when in fact he comes from π_j is given by

$$p_{ij} = \int \phi_i(x) L_j(x) dx. \quad \text{--- (4)}$$

In particular, if an individual is in fact from π_i the prob. of correct allocation is

$$p_{ii} = \int \phi_i(x) L_i(x) dx.$$

Defⁿ: Say that one discrimination rule d with probabilities of correct allocation $\{p_{ii}\}$ is as good as another rule d' with probabilities $\{p'_{ii}\}$ if

$$p_{ii} \geq p'_{ii} \quad \text{for } i = 1, \dots, g.$$

Say that d is better than d' if at least one of the inequality is strict.

If d is a rule for which there is no better rule, say that d is **admissible**.

Theorem: All Bayes discrimination rules (including the ML rule) are admissible.

Proof: Let d^* denote the Bayes rule with respect to prior prob. π . Suppose there exists a rule d which is better than this rule. Let $\{p_{ij}\}$ and $\{p_{ij}^*\}$ denote the probabilities of correct classification for this rule and the Bayes rule, respectively. Then because d is better than d^* and since $\pi_j > 0$ for j ,

$$\sum \pi_i p_{ii} > \sum \pi_i p_{ii}^*.$$

However using ③ and ④

$$\begin{aligned} \sum \pi_i p_{ii} &= \sum_i \int \phi_i(x) L_i(x) dx \\ &\leq \sum_i \int \phi_i(x) \max_j L_j(x) dx \end{aligned}$$

$$\begin{aligned}
&= \int \left[\sum_i \phi_i(x) \right] \max_j L_j(x) dx \\
&= \int \max_j \pi_j L_j(x) dx \\
&= \int \sum \phi_i^*(x) \pi_i L_i(x) dx \\
&= \sum \pi_i p_{ii}^*
\end{aligned}$$

which contradicts the above statement. Hence the theorem is proved.

Decision theory: The discrimination problem can be phrased in the language of decision theory. Let

$$K(i, j) = \begin{cases} 0, & i = j \\ c_{ij}, & i \neq j. \end{cases} \quad \text{--- (5)}$$

be a loss function representing the cost or loss incurred when an obj is allocated to π_i when in fact it comes from π_j .

If d is a discr. rule with allocation prob. $\phi_i(x)$ then the risk function is defined by

$$\begin{aligned}
 R(d, j) &= E(K(d(x), j) | \pi_j) \\
 &= \sum_i K(i, j) \int \phi_i(x) L_j(x) dx \quad \text{--- (6)} \\
 &= \sum_i c_{ij} p_{ij}
 \end{aligned}$$

and represents the expected loss given that the observation comes from π_j .

In particular if $c_{ij} = 1 \ \forall i \neq j$, then

$$R(d, j) = 1 - p_{jj}$$

represents the misallocation probabilities.

Defⁿ: Bayes Risk: If prior prob. exist then the Bayes risk can be defined by

$$r(d, \pi) = \sum_j \pi_j R(d, j)$$

Say that a decis. rule d is "**admissible**" if there exists no other rule d' such that $R(d', j) \leq R(d, j)$ for all j , with at least one strict inequality.

Discrimination under Estimation:

The Sample Discrimination Rule:

The sample ML discrimination rule is useful when the forms of the distributions of $\pi_1, \dots, \pi_g$ are known, but their parameters must be estimated from a data matrix $X(n \times p)$.

$X^T = (X_1^T, \dots, X_g^T)$ and that X_i contains n_i observations from π_i .

For example, suppose the groups are assumed to be samples from the multivariate normal distribution with different means and the same covariance matrix.

Let $\bar{x}_i$ and S_i denote the sample mean and covariance matrix of the i -th group. The unbiased estimates of $\mu_1, \dots, \mu_g$ and Σ are $\bar{x}_1, \dots, \bar{x}_g$ and

$S_g = \sum n_i S_i / (n - g)$. The sample ML discriminant rule is then obtained by inserting these estimates in Theorem 11. In particular when $g=2$ the sample ML discriminant rule allocates x to π_1 if and only if

$$Q^T \left\{ X - \frac{1}{2}(X_1 + X_2) \right\} > 0 \quad \text{--- (I)}$$

where $Q = S_u^{-1}(\bar{X}_1 - \bar{X}_2)$.

Example: Let us start with Brown mean and variance.

$$\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mu_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mu_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$$

M1 allocation rule:

$$\Sigma^{-1} = \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix}$$

$$(\mu_1 - \mu_2) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad \Sigma^{-1}(\mu_1 - \mu_2) = \begin{bmatrix} 2 \\ -5 \end{bmatrix} = \alpha_{12}$$

$$\mu_1 + \mu_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \alpha_{12}^T \left(X - \frac{1}{2}(\mu_1 + \mu_2) \right) &= (2, -5) \begin{bmatrix} x_1 \\ x_2 + 1/2 \end{bmatrix} \\ &= 2x_1 - 5x_2 + 5/2. \quad \text{--- (i)} \end{aligned}$$

$$\mu_1 - \mu_3 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad \Sigma^{-1}(\mu_1 - \mu_3) = \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \alpha_{13}$$

$$(\mu_1 + \mu_3) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\alpha_{13}^T \left(\tilde{x} - \frac{1}{2} (\mu_1 + \mu_3) \right) = [-1, 2] \begin{bmatrix} x_1 - 1/2 \\ x_2 \end{bmatrix}$$

$$= -x_1 + 2x_2 + 1/2 \quad \text{--- (i')}$$

$$\mu_2 - \mu_3 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad \Sigma^{-1}(\mu_2 - \mu_3) = \begin{bmatrix} -3 \\ 7 \end{bmatrix} = \alpha_{23}$$

$$\alpha_{23}^T \left(\tilde{x} - \frac{1}{2} (\mu_2 + \mu_3) \right) = [-3, 7] \begin{bmatrix} x_1 - 1/2 \\ x_2 - 1/2 \end{bmatrix}$$

$$= -3x_1 + 7x_2 - 2. \quad \text{--- (ii')}$$

$$2x_1 - 5x_2 + 5/2 = -x_1 + 2x_2 + 1/2$$

$$3x_1 - 7x_2 = -2 \quad \text{--- (1')}$$

$$-x_1 + 2x_2 + 1/2 = -3x_1 + 7x_2 - 2$$

$$2x_1 - 5x_2 = -5/2 \quad \text{--- (2')}$$

$$2 \times (1') - 3 \times (2')$$

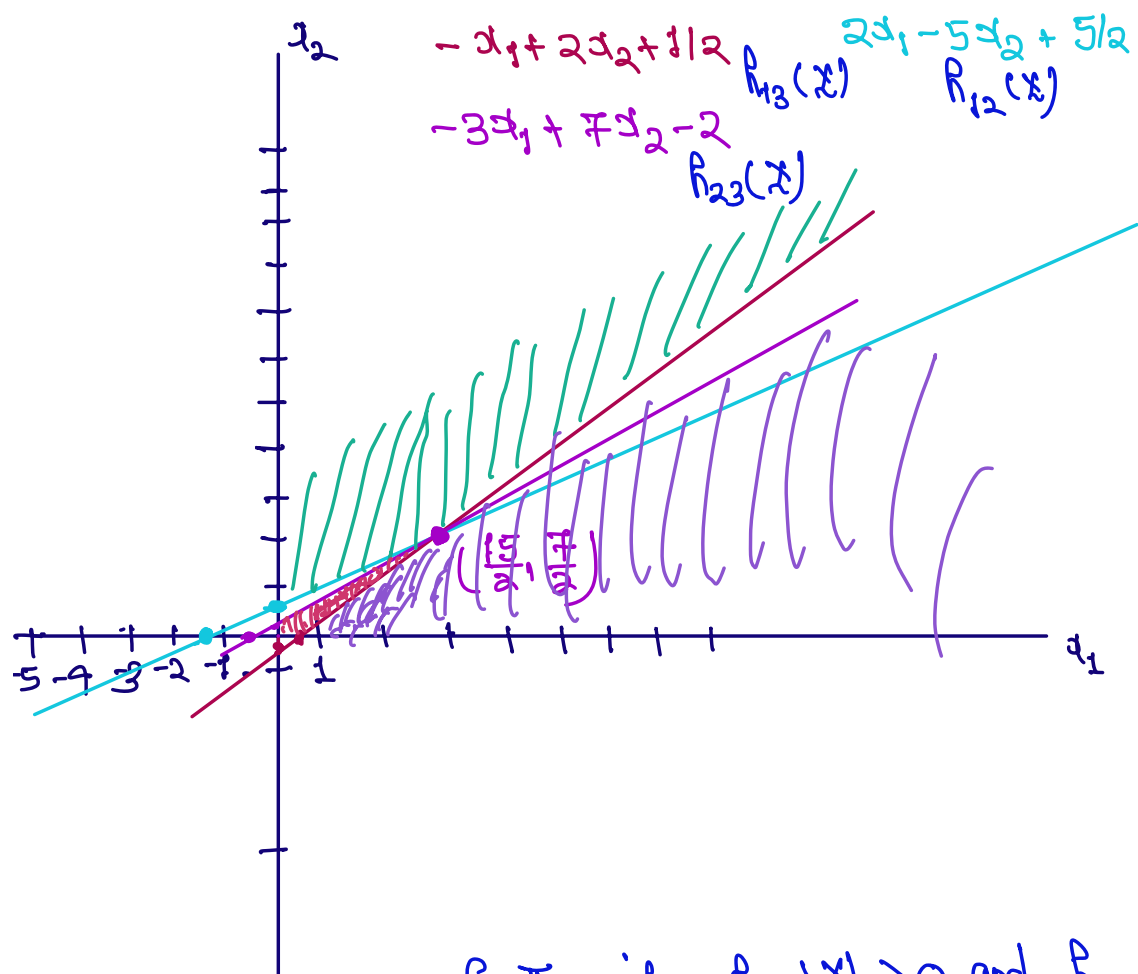
$$6x_1 - 14x_2 = -4$$

$$\begin{array}{r} 6x_1 - 15x_2 = -15/2 \\ - \quad \quad \quad \uparrow \\ \hline \end{array}$$

$$\Rightarrow \quad x_2 = \frac{15}{2} - 4 = \frac{7}{2}.$$

$$3x_1 - \frac{49}{2} = -2$$

$$3x_1 = \frac{45}{2} \Rightarrow x_1 = \frac{15}{2}$$



Allocate x to

$$\begin{cases} \pi_1 & \text{if } R_{12}(x) > 0 \text{ and } R_{13} > 0 \\ \pi_2 & \text{if } R_{12}(x) < 0 \text{ and } R_{23} > 0 \\ \pi_3 & \text{if } R_{13}(x) < 0 \text{ and } R_{23} < 0 \end{cases}$$

For unknown μ_i and Σ , insert associated estimates.

The likelihood ratio discrimination rule:

An alternative rule to the sample ML allocation rule uses the "likelihood ratio criterion". This rule involves calculating the likelihoods of the hypotheses

H_i : x and the rows of X_i are from π_i and the rows of X_j are from π_j $j \neq i$,

for $i=1, \dots, g$. Then x is allocated to the group for which the hypothesis H_i has the largest likelihood.

Consider the discrimination problem between $g=2$ multivariate populations with the same covariance matrix.

Derivation of MLEs of μ_1, μ_2 and Σ under H_1 :

$$\text{Let } x_1^{(1)}, \dots, x_{n_1}^{(1)} \sim N(\mu_1, \Sigma) \\ x_1^{(2)}, \dots, x_{n_2}^{(2)} \sim N(\mu_2, \Sigma).$$

Likelihood,

$$L(\mu_1, \mu_2, \Sigma) = \frac{1}{(2\pi)^{\frac{n_1+n_2+1}{2}} |\Sigma|^{\frac{n_1+n_2+1}{2}}} \cdot \exp \left\{ - \sum_{i=1}^{n_1} (x_i^{(1)} - \mu_1)^T \Sigma^{-1} (x_i^{(1)} - \mu_1) - \sum_{j=1}^{n_2} (x_j^{(2)} - \mu_2)^T \Sigma^{-1} (x_j^{(2)} - \mu_2) \right\}$$

$$\log L = C - (n_1 + n_2 + 1) \log |\Sigma| - \sum_{i=1}^{n_1} (x_i^{(1)} - \mu_1)^T \Sigma^{-1} (x_i^{(1)} - \mu_1) - \sum_{j=1}^{n_2} (x_j^{(2)} - \mu_2)^T \Sigma^{-1} (x_j^{(2)} - \mu_2)$$

$$\hat{\mu}_1 = \frac{1}{n_1+1} \sum_{i=0}^{n_1} x_i^{(1)}, \quad \bar{x}^{(1)} = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i^{(1)}$$

$$= \frac{x + n_1 \bar{x}^{(1)}}{n_1+1}.$$

$$\hat{\mu}_2 = \bar{x}^{(2)}.$$

$$\hat{\Sigma}_1 = \frac{1}{n_1+n_2+1} \left[\sum_{i=1}^{n_1} (x_i^{(1)} - \hat{\mu}_1) (x_i^{(1)} - \hat{\mu}_1)^T \right. \\ \left. + (x - \hat{\mu}_1) (x - \hat{\mu}_1)^T \right. \\ \left. + \sum_{j=1}^{n_2} (x_j^{(2)} - \bar{x}^{(2)}) (x_j^{(2)} - \bar{x}^{(2)})^T \right]$$

$$\sum_{i=1}^{n_1} (x_i^{(1)} - \hat{\mu}_1) (x_i^{(1)} - \hat{\mu}_1)^T + (x - \hat{\mu}_1) (x - \hat{\mu}_1)^T$$

$+ \bar{x}^{(1)} - \bar{x}^{(1)} \quad + \bar{x}^{(1)} - \bar{x}^{(1)}$

$$= \sum (x_i^{(1)} - \bar{x}^{(1)}) (x_i^{(1)} - \bar{x}^{(1)})^T + n_1 (\bar{x}^{(1)} - \hat{\mu}_1) (\bar{x}^{(1)} - \hat{\mu}_1)^T$$

$$+ (x - \hat{\mu}_1) (x - \hat{\mu}_1)^T$$

$$\bar{x}^{(1)} - \hat{\mu}_1 = \frac{n_1 \bar{x}^{(1)} + \bar{x}^{(1)} - n_1 \bar{x}^{(1)} - x}{n_1+1}$$

$$= \frac{\bar{x}^{(1)} - x}{n_1+1}$$

$$\alpha - \hat{\mu}_1 = \frac{n_1 \alpha + 1 - n_1 \bar{\alpha}^{(1)} - 1}{n_1 + 1} = \frac{n_1}{(n_1 + 1)} (\alpha - \bar{\alpha}^{(1)})$$

$$\Rightarrow \sum (x_i^{(1)} - \bar{\alpha}^{(1)}) ()^T + \left[\frac{n_1}{(n_1 + 1)^2} + \frac{n_1^2}{(n_1 + 1)^2} \right] (\alpha - \bar{\alpha}^{(1)}) ()^T$$

$$= \sum (x_i^{(1)} - \bar{\alpha}^{(1)}) + \frac{n_1}{(n_1 + 1)} (\alpha - \bar{\alpha}^{(1)}) ()^T$$

$$\Rightarrow \hat{\Sigma}_1 = \frac{1}{n_1 + n_2 + 1} \left[A + \frac{n_1}{(n_1 + 1)} (\alpha - \bar{\alpha}^{(1)}) ()^T \right]$$

Similarly under H_2 ,

$$\hat{\mu}_1 = \bar{\alpha}^{(1)}$$

$$\hat{\mu}_2 = \frac{n_2 \bar{\alpha}^{(2)} + 1}{n_2 + 1}$$

$$\hat{\Sigma}_2 = \frac{1}{n_1 + n_2 + 1} \left[A + \frac{n_2}{n_2 + 1} (\alpha - \bar{\alpha}^{(2)}) ()^T \right]$$

The likelihood ratio criterion is, therefore, the $(n_1 + n_2 + 1)/2$ th power of

$$\frac{|\hat{\Sigma}_2|}{|\hat{\Sigma}_1|} = \frac{A + \frac{n_2}{n_2 + 1} (\alpha - \bar{x}^{(2)}) (\quad)^T}{A + \frac{n_1}{n_1 + 1} (\alpha - \bar{x}^{(1)}) (\quad)^T}$$

$$= \frac{1 + \frac{n_2}{n_2 + 1} (\alpha - \bar{x}^{(2)}) A^{-1} (\quad)^T}{1 + \frac{n_1}{n_1 + 1} (\alpha - \bar{x}^{(1)}) A^{-1} (\quad)^T}$$

We accept H_1 , and allocate x to π_1 if and only if

$$\frac{n_2}{1 + n_2} (\alpha - \bar{x}^{(2)})^T A^{-1} (\alpha - \bar{x}^{(2)})$$

$$> \frac{n_1}{1 + n_1} (\alpha - \bar{x}^{(1)})^T A^{-1} (\alpha - \bar{x}^{(1)})$$

In the special case when $n_1 = n_2 = n$, this likelihood criterion is equivalent to the sample ML discriminant rule.

Multi-sample Hypothesis:

$$H_0: \mu_1 = \dots = \mu_K \text{ given that } \Sigma_1 = \dots = \Sigma_K$$

$$H_1: \mu_i \neq \mu_j \text{ for at least a pair } (i, j) \\ \text{given that } \Sigma_1 = \dots = \Sigma_K.$$

Under H_0 the MLE of μ and Σ are

$$\hat{\mu} = \bar{X}, \quad \hat{\Sigma} = \frac{1}{n} \sum (X_i - \bar{X})(X_i - \bar{X})^T \\ = \frac{T}{n}.$$

Under H_1 , $\hat{\mu}_i = \bar{X}_i$ and $\hat{\Sigma} = \frac{W}{n}$ where

$$W = \sum_{i=1}^K n_i S_i$$

The LRT is given by

$$\lambda = \left\{ |W| / |TS| \right\}^{n/2} = |T^{-1}W|^{n/2}$$

where $T = nS$ is the "total sum of square" matrix.

$$B = T - W = \sum n_i (\bar{X}_i - \bar{X})(\bar{X}_i - \bar{X})^T$$

is "between-groups" SS matrix. W is "within-group"

SS matrix.

$$\lambda^{2/n} = \frac{|W|}{|B+W|} = |I + W^{-1}B|^{-1} \\ \sim \Lambda(p, n-k, k-1)$$

is Wilk's Λ statistic under H_0 .

Defⁿ: When $A \sim W_p(I, m)$ and $B \sim W_p(I, n)$ are independent, $m \geq p$, we say that

$$\Lambda = \frac{|A|}{|A+B|} = |I + A^{-1}B|^{-1} \sim \Lambda(p, m, n)$$

has a Wilk's lambda distribution with parameters p, m and n .

It can be show that

$$\left. \begin{array}{l} W \sim W_p(\Sigma, n-k) \\ B \sim W_p(\Sigma, k-1) \end{array} \right\} \text{ indep.}$$

Let us come back to discrimination analysis!

Is Discrimination Worthwhile?

To check whether or not the discriminant analysis is worthwhile, it is interesting to test the hypothesis $\mu_1 = \dots = \mu_g$ given $\Sigma_1 = \dots = \Sigma_g$.

The above hypothesis can be tested using Wilk's Λ statistics.

Fisher's Linear Discriminant Function:

Fisher suggested to look for the linear function $\mathbf{a}^T \underline{x}$ which maximized the ratio of the between-groups sum of squares to the within-groups sum of squares; that is let

$$\underline{y} = \mathbf{X} \mathbf{a} = \begin{bmatrix} x_{1a} \\ \vdots \\ x_{ga} \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_g \end{bmatrix}$$

be a linear combination of the columns of $\mathbf{X}$. Then $\underline{y}$ has total sum of squares

$$\underline{y}^T \mathbf{H} \underline{y} = \mathbf{a}^T \mathbf{X}^T \mathbf{H} \mathbf{X} \mathbf{a} = \mathbf{a}^T \mathbf{T} \mathbf{a},$$

which can be partitioned as a sum of the within-group sum of squares,

$$\sum_{i=1}^g \tilde{y}_i^T H_i \tilde{y}_i = \sum a^T x_i^T H_i x_i a$$

$$= a^T W a,$$

plus the between-group sum of squares

$$\sum n_i (\bar{y}_i - \bar{y})^2 = \sum n_i \{ a^T \bar{x}_i - \bar{x} \}^2$$

$$= a^T B a$$

where $\bar{y}_i$ is the mean of the i th sub-vector $\tilde{y}_i$ of $\tilde{y}$ and H_i is $n_i \times n_i$ centering matrix given by

$$H_i = I_i - \frac{1}{n_i} \mathbf{1}_{n_i} \mathbf{1}_{n_i}^T$$

$$C_1 = \sum H_i, \quad C_2 = \sum n_i^{-1} \mathbf{1}_i \mathbf{1}_i^T - n^{-1} \mathbf{1} \mathbf{1}^T$$

$$W = X^T C_1 X, \quad B = X^T C_2 X.$$

The ratio of between-group sum of squares to within-group sum of squares is

$$\frac{q^T B q}{q^T W q}$$

—— (1)

If q is a vector which minimizes (1) we shall call the linear function $q^T x$, Fisher's linear discrimination function on the first canonical variate.

Theorem: The vector q in Fisher's linear discriminant function is the eigenvector of $W^{-1}B$ corresponding to the largest eigenvalue.

Proof:

Result: Let A and B be two symmetric matrices. Suppose that $B > 0$. Then the maximum (minimum) of $\frac{x^T A x}{x^T B x}$ given

$$x^T B x = 1$$

is attained when x is the eigenvector of $B^{-1}A$ corresponding to the largest (smallest) eigenvalue of $B^{-1}A$.

Once the linear discrimination function has been calculated, then allocate x to that population whose mean score is closest to $q^T x$, that is allocate x to π_j if

$$|a_2^T x - a_2^T \bar{x}_j| < |a^T x - a^T \bar{x}_i|$$

for all $i \neq j$.

Rank $B = 1$ and

$$B = \frac{n_1 n_2}{n} d d^T$$

where $d = (\bar{x}_1 - \bar{x}_2)$

Thus $W^{-1}B$ has only one non-zero eigen value that is

$$\text{tr } W^{-1}B = \frac{n_1 n_2}{n} d^T W^{-1}d$$

$$W^{-1}B v = \lambda v, \text{ let } v = a = W^{-1}d$$

$$\lambda^* \underbrace{W^{-1} d d^T \cdot W^{-1} d}_{\substack{\text{tr } W^{-1}B \\ \lambda^*}} = \lambda^* d^T W^{-1} d \cdot \underbrace{W^{-1} d}_a$$

$$\Rightarrow a = W^{-1}d$$

Then the discriminant rule becomes

allocate x to π_1 if $d^T W^{-1}(x - \frac{1}{2}(\bar{x}_1 + \bar{x}_2)) > 0$

This is same as sample allocation rule.

Probability of Misclassification: Estimates of p_{12} can be obtained as follows.

Example: $N_p(\mu_1, \Sigma)$ $N_p(\mu_2, \Sigma)$

If $\mu = \frac{1}{2}(\mu_1 + \mu_2)$, then when x comes from π_1 .

$$\alpha^T(x - \mu) \sim N[\alpha^T(\mu_1 - \mu), \alpha^T \Sigma \alpha] \\ = N[\frac{1}{2}\alpha^T(\mu_1 - \mu_2), \alpha^T \Sigma \alpha]$$

with $\alpha = \Sigma^{-1}(\mu_1 - \mu_2)$, when x comes from π_1

$$R(x) = \alpha^T(x - \mu) \sim N[\frac{1}{2}\Delta^2, \Delta^2] \text{ where}$$

$$\Delta^2 = (\mu_1 - \mu_2)^T \Sigma^{-1}(\mu_1 - \mu_2)$$

Similarly when x comes from π_2 then

$$R(x) \sim N[-\frac{1}{2}\Delta^2, \Delta^2]$$

Thus the prob of misclassifications are.

$$p_{12} = P[R(x) > 0 | \pi_2]$$

$$= \Phi\left(-\frac{E(R)}{\sqrt{\text{var}(R)}}\right) = \Phi\left(-\frac{1}{2}\Delta\right)$$

where Φ is the standard normal distribution function. Similarly $p_{21} = \Phi\left(-\frac{1}{2}\Delta\right)$.

an estimate of Δ^2 is

$$D^2 = (\bar{X}_1 - \bar{X}_2)^T S_n^{-1} (\bar{X}_1 - \bar{X}_2).$$